BIOT6900 – AI-Powered Biotech Workflows

Module 2 Assignment — Computational Lab 2

Chosen Disease: Ovarian Carcinoma

A. Disease & data choice

1. Why this disease?

The reason I chose ovarian carcinoma is because it is one of the most common gynecological cancers and is very well characterized CPTAC proteogenomic cohort on LinkedOmics. High grade serous ovarian carcinoma (HGSOC) alone makes for 70% of the ovarian cancer incidents and is highly deadly in gynecology. I found it interesting and think that further study on target discovery could be helpful in the future of this project.

2. What datasets did you use, from where, and were they open or gated?

Following are the layers and the sources I used:

1. **Transcriptomics (RNA):** CPTAC-OV RNAseq from LinkedOmics. There are 82 tumor samples measured by Illumina HiSeq at the gene level (log2FPKM values). One thing I found interesting was that there are no matched normal RNA samples, the reason being normal ovarian tissue is hard to get and so wasn't included in this cohort. I have noted this as a limitation and have explained how I worked around it in the following parts.
2. **Proteomics (protein):** CPTAC-OV global proteome from LinkedOmics, measured using TMT mass spectrometry at gene level. This layer uses both tumor (83 samples) and normal (20 samples). Thus I followed the assignment guidelines for proper tumor vs normal differential using Welch's t-test.
3. **Genomics (GWAS):** GWAS catalog associations for ovarian carcinoma (MONDO_0005140, with child traits included capturing subtypes like serous, endometrioid etc.). I got 732 raw associations which was processed to 557 unique genes having strongest -log10(p) values.

All datasets used are open access. RNA and protein are from the same CPTAC-OV study (Zhang et al., 2016), so patients are matched in these layers.

3. Matched or unmatched across layers?

The GWAS data is population-level germline data from separate studies. Everything is integrated at the gene level using inner join on gene symbols. Thus I had 154 genes appearing across all the three layers.

B. Top targets

1. Check your top hits against known biology for your disease (use Open Targets or the literature).

I got 154 genes after the 3-way join, the concordance rate was 59.7% (92 of 154 genes showed RNA and protein in same direction). Following are two of my top hits and why they might be significant:

1. **FGFR2 (rank 1, score 0.530):** Fibroblast growth factor receptor 2 ranked first and showed a very strong GWAS signal (neglog10p = 253). FGFR2 is one of the most well-established loci in ovarian cancer and is replicated consistently across multiple GWAS. Another interesting point I noticed was that the RNA was slightly higher (+1.87) while protein is slightly below (-0.43)
2. **HSPB7 (rank2, score 0.412):** Heat shock protein beta 7 was second. It showed downregulation of both RNA (-1.25) and protein (-2.22) which is supported by literature showing it is concordant. HSPB7 is reportedly a tumor suppressor in many cancer types, seeing that it is downregulated in both layers shows it is an interesting target for drug candidates.

2. Which known genes did you recover?

Other characteristic genes of ovarian carcinoma I noted were PAX8 (rank 25, score 0.184) and TP53 (rank 66, score 0.109) which I found to be interesting as I expected them to be higher ranking. A possible reason for slightly lower rank of TP53 could be that it is somatic and not germline, hence GWAS didn't pick it up strongly.

3. Any non-obvious hit worth a second look?

I think SYNPO2 (rank 3, score 0.410) is an interesting hit. Synaptopodin 2 is strongly downregulated at both RNA (-0.28) and protein (-2.43). What I found interesting is that it is not a classic ovarian cancer gene but is actually associated with actin proteins working in cell migration and cytoskeletal regulation. It's downregulation is interesting as it might suggest a strong role in loss of normal tissue structure as the cancer progresses posing as interesting topic for further studies.

C. Read a discordant gene (if you have RNA + protein) — what biology could explain the disagreement?

Altogether there were 13 discordant genes in the top 50. One of them is LMNA (Lamin A/C) with a rank of 13 and score of 0.231. The RNA was strongly upregulated (+3.81) while protein was downregulated (-0.65). I think there could be two possible reasons for this:

1. **Post-translational degradation:** tumor cells transcribe LMNA at higher levels but the protein is getting degraded faster than it is made
2. **RNA measure isn't a good comparison of tumor vs normal:** since normal RNA samples are missing, it could be that this positive RNA value shows LMNA is above the cohort average but not actually upregulated when compared to normal tissues.

D. What do your specific datasets and integration level (matched vs. gene-level) let you claim, and not claim?

a. Using Custom Weights

I assigned a custom weights of: RNA = 0.20, Protein = 0.40, GWAS = 0.40. The main support for this is that the RNA layer is weaker for this analysis as there are no normal dataset available hence real tumor vs normal fold change could not be computed. Protein on the other hand has a proper tumor vs normal comparison (83 vs 20 samples) in the Welch's t-test thus making it more reliable. GWAS shows germline causal risk variants from large population studies thus making it more reliable as well.

b. Limitations

1. No matching normal RNA: To work around this, I considered the transcriptomic effect to be deviated from the cohort mean rather than tumor vs normal log2FC. As compensation I have decreased the weight for the RNA layer as genes that are discordant could actually be concordant if there was normal RNA data

2. Small normal protein sample: The protein layer has 83 tumors but only 20 normals, this affects the statistics especially in genes with high variance

3. Small surviving gene set: only 154 genes survived the three way inner join out of more than 10,000 protein coding genes. This is due to GWAS covering only few hundred genes that are significantly associated for ovarian cancer. There could be other biologically relevant genes that were not picked up in GWAS.

E. References

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